

3/11/85

Stuart,

Outline of work for the next couple of weeks!

Project I

- 1) Read section 4.10 of my thesis. [Ask me for copies of relevant papers: Haskings, Dales and Read, Moore.]
- 2) Examine effect on parameter estimates/quantile estimates (+ errors) of accounting for dependence in the margins when
 - (i) no per margin parameters are common (are margin parameters π indep for one margin to another)
 - (ii) k is same for each margin (are (μ_1, τ) indep of (μ_2, τ_2))
 - ~~Do this~~ (iii) μ, τ have a covariate relationship.

Do this only in the bivariate case using a simulation study. I have programs that estimate bivariate extreme value dist (for exponential margins) and somewhere have a prog. to simulate bivariate e.v. d. (for exponential margins).

- 3) Compare results with Haskings concepts and those in Sect. 4.10.
- 4) Obtain joint quantile estimates with s.e.

Project II

- 1) Read de Haan & Reniers (77), de Haan (1985) and sections 3.14 and 4.9, ^{also} section 4.12 ^{all chapter 4 if not already read.}
note on p171 L-5 ~~should read and~~ change
'that exceed u_n ' for 'for which the L norm exceeds u_n '

- 2) Start to think how to combine this approach with marginal estimation to obtain closed form expressions in the original variables (ie as in project I). Is this possible or is it even needed. - may depend on Project I results.

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- 3) See if there is any connection with concomitants.

Avoid data for the moment!!

Jonathan